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Analysis of seismic sequences by using the method of visibility graph

L. Telesca[1(a)] and M. Lovallo[2]

[1] *National Research Council, Institute of Methodologies for Environmental Analysis - C.da S.Loja, 85050 Tito, Italy, EU*
[2] *ARPAB - 85100, Potenza, Italy, EU*

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Abstract – The seismicity of Italy between April 16, 2005 and December 31, 2010 is investigated by means of the visibility graph method. The degree distribution of the magnitude point process is power-law–shaped. The method does not seem to detect time-clustering structures in the magnitude point process. The degree distribution does not change significantly with the increase of the magnitude threshold from 1.9 to 3.5. The exponent of the degree distribution shows a certain stability *vs.* the threshold magnitude, ranging between 3.00 and 3.25, with mean value 3.11 ± 0.06.

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The analysis of seismic sequences as a time series has always been representing one of the most debated questions in geophysics. The representation of a seismic sequence as a time series allows to apply techniques derived from the nonlinear analysis, by which fundamental properties in terms of clustering, fractality, correlation, memory can be identified and quantified. A correlated structure in the time occurrences of earthquakes [1] and in the seismic interevent distance time series [2] was revealed by using the Allan factor (AF) (a method able to detect time clustering in point processes [3]) and the detrended fluctuation analysis (DFA) (a method able to detect correlations in nonstationary time series [4]). Long-range correlations were also revealed in the couples ($\Delta\tau$, Δs), where $\Delta\tau$ and Δs are, respectively, the interevent time and distance between successive earthquakes [5]. A nontrivial relationship involving time clustering, magnitude and spatial location of earthquakes represented as time series was shown in several seismic areas [2]. Very recently, new approaches based on Tsallis entropy have investigated the nonextensive properties of the earthquake magnitude distribution [6–8], disclosing relationships between the Gutenberg-Richter b-value and the nonextensive parameter q [9].

After the seminal paper by Lacasa *et al.* [10], an increasing interest has been devoted to the method of *visibility graph* (VG), which has revealed its potential in describing the main characteristics of a time series. This method maps a time series into a network, which reflects several properties of the time series. In turn, the investigation of a network constructed from the time series through the VG method can disclose nontrivial information about the time series itself.

The scheme of the VG method is illustrated in fig. 1, which shows the first 20 values of the series of magnitudes of earthquakes occurred in Italy from April 16, 2005 to December 31, 2010; the data are indicated by vertical arrows. Each arrow is linked with all those that can be seen from the top of the considered one (dotted lines). As a graph, each arrow (that is a value of the time series) is a node, and two nodes are connected if visibility exists between the corresponding arrows; visibility, in this case, means that there exists a straight line connecting the series data, but not intersecting any intermediate arrow.

The visibility criterion [10] can be defined as follows: two arbitrary data values (t_a, y_a) and (t_b, y_b) are visible to each other if any other data $(t_c,\ y_c)$ placed between them fulfills the following constraint:

$$y_c < y_b + (y_a - y_b)\,\frac{t_b - t_c}{t_b - t_a}. \tag{1}$$

The following properties always hold [10]: 1) connection: each node is visible at least by its nearest neighbors (left and right); 2) absence of directionality: no direction is defined in the links; 3) invariance under affine

(a) E-mail: luciano.telesca@imaa.cnr.it

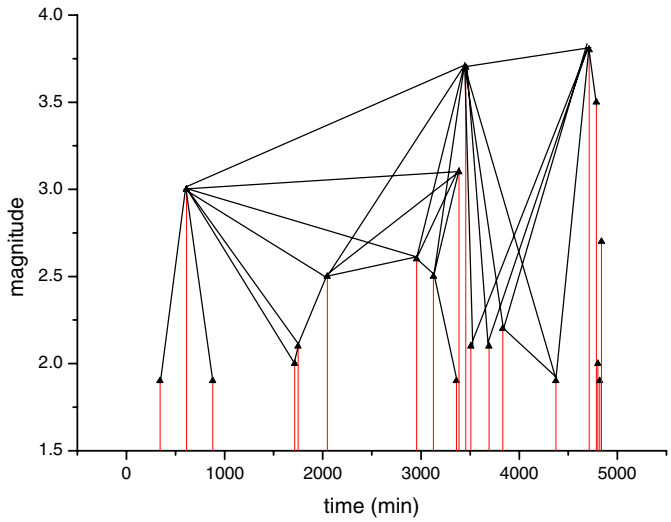

Fig. 1: (Colour on-line) The first 20 magnitude data (red vertical arrows) of the Italian seismic sequence from April 16, 2005 to December 31, 2010. The visibility rays (black lines) between the data define the links connecting the nodes.

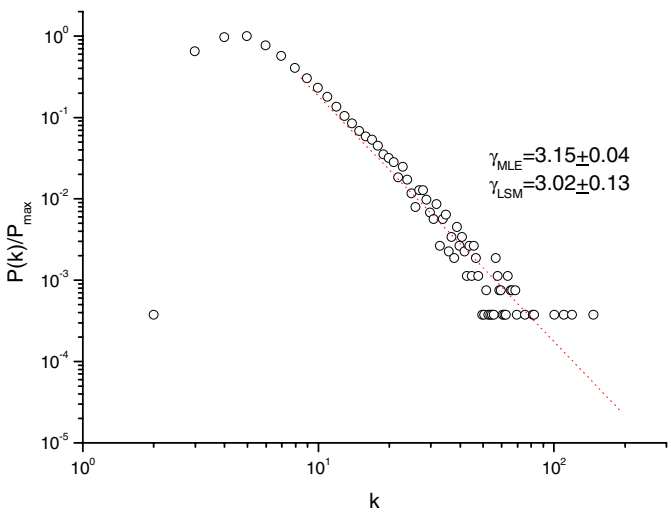

Fig. 2: (Colour on-line) Degree distribution of the visibility graph associated with the magnitude series of the Italian earthquakes occurred from April 16, 2005 to December 31, 2010 for threshold magnitude $m_T = 1.9$. The distribution is not normalized. The degree γ calculated by the LSM and the MLE is, respectively, 3.02 ± 0.13 and 3.15 ± 0.04. The red dotted line represents the line fitting the distribution plotted in log-log scales through the LSM.

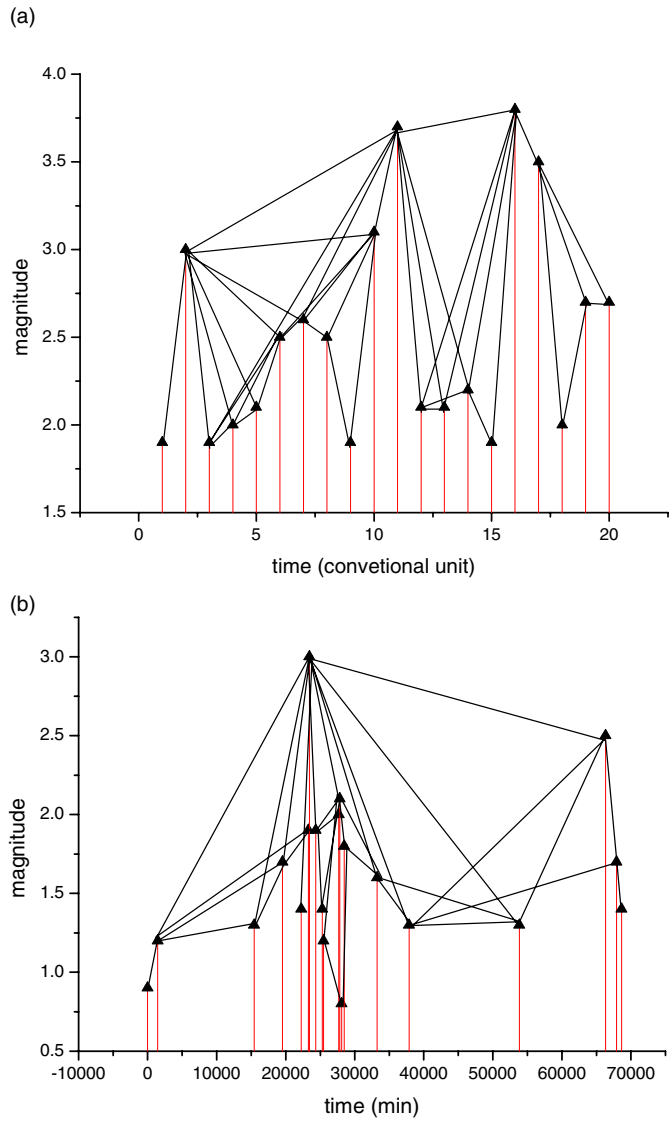

Fig. 3: (Colour on-line) (a) The first 20 magnitude data (red vertical arrows) of the Italian seismic sequence from April 16, 2005 to December 31, 2010. The original occurrence times are substituted by regular (conventional unit). (b) The first 20 magnitude data (red vertical arrows) of the randomly shuffled sequence of seismic events occurred in Italy from April 16, 2005 to December 31, 2010.

transformations (rescaling of both axes and horizontal and vertical translations) of the time series.

It was shown that the graph constructed on the base of the VG method saves the main properties of a time series: periodic time series are converted into regular graphs, random time series into random graphs, and fractal time series into scale-free networks [10–12].

Recently Lacasa *et al.* [13] have shown that the VG derived from a fractional Brownian motion (fBm) is scale invariant, with a power-law distribution of the degree distribution, $P(k) \sim k^{-\gamma}$, where k is the degree of a specified node, and γ is the power-law exponent linearly related with the Hurst exponent H of the fBm, $\gamma = 3 - 2H$. Due to the relationship between the H and the spectral exponent β of a $1/f^{\beta}$ noise, it was shown that the VG method allows to detect long-range dependencies through the relationship $\gamma = 4 - \beta$ [13].

In this letter we apply the VG method to the sequence of magnitudes of seismic events occurred in Italy between April 16, 2005 to December 31, 2010 (data extracted from [14]). The completeness magnitude of the investigated catalogue is 1.9. The magnitude time series is

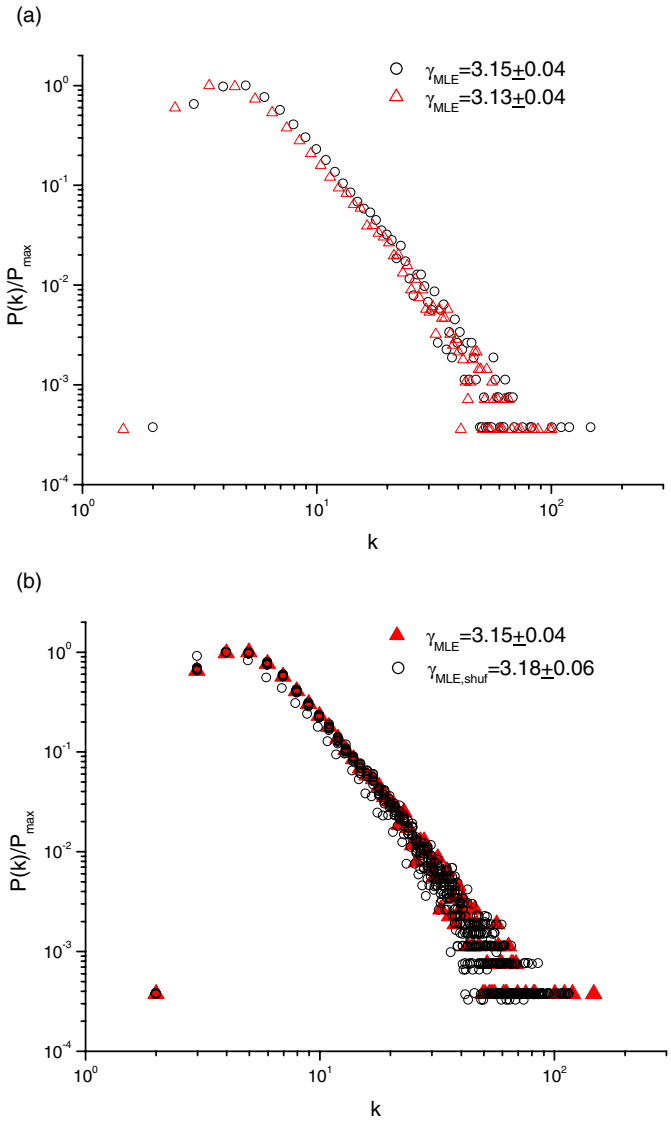

Fig. 4: (Colour on-line) (a) Comparison between the degree distributions of the visibility graph associated with the original magnitude series (black circles) and the conventional one (red triangles). The exponents in both cases are very close and the degree distributions almost overlapping. (b) Comparison between the degree distributions of the visibility graph associated with the original magnitude series (red triangles) and the shuffled ones (black circles). The exponents in both cases are almost identical.

a marked temporal point process, described by sum of Dirac's delta centered on the occurrence time with amplitude proportional to the magnitude of the event; fig. 1 shows the representation of the first twenty earthquakes of the whole analysed sequence. Figure 2 shows the degree distribution in log-log scales of the VG for the magnitude series with 1.9 as threshold magnitude m_T. The power law is visible for $k \geqslant 10$. We calculated the exponent γ: i) by fitting the degree distribution plotted in log-log scales in the linear range using the linear least square method (LSM), and ii) by using the maximum likelihood

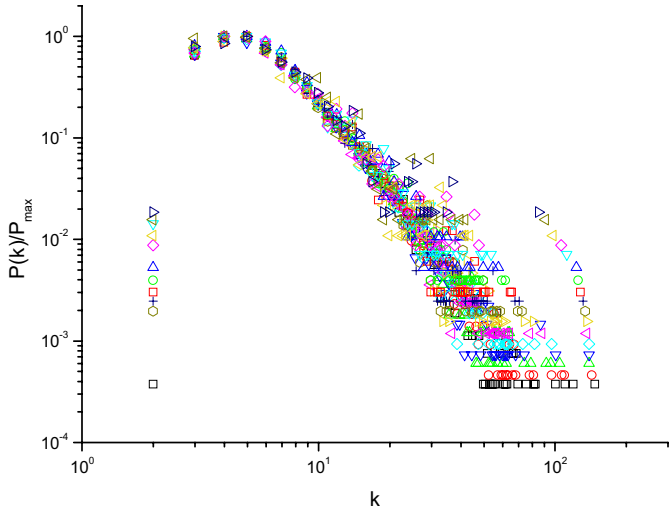

Fig. 5: (Colour on-line) Degree distributions of the visibility graphs associated with the magnitude series for threshold magnitudes m_T ranging from 1.9 to 3.5. The distributions are normalized to their maximum. There is a collapsing effect of all the distributions, suggesting a threshold magnitude-invariant behaviour.

estimation (MLE). Regarding the MLE, we used the following equation [15]:

$$\gamma_{MLE} = 1 + n \left[\sum_{i}^{n} \ln \frac{k_i}{k_{min} - 0.5} \right], \tag{2}$$

where k_{min} is the smallest value of the node k for which the power law holds and n is the number of values $k \geqslant k_{min}$. The estimation error is given by [15]

$$\sigma_{MLE} = \frac{y_{MLE} - 1}{\sqrt{n}}. \tag{3}$$

In our case $k_{min} = 10$ and $n = 2697$. The values for the exponent γ is 3.02 ± 0.13 (LSM) and 3.15 ± 0.04 (MLE). The two values of the slopes obtained through the LSM and the MLE methods are almost identical. On the base of the relationship $\gamma = 4 - \beta$ [13], which, however, has been obtained for fBm processes, we can argue that the magnitude series is long-range–correlated. The power-law behaviour of the degree distribution is a fingerprint of the fractality of the magnitude sequence. The heavy-tailed degree distribution is mostly due to the events (fewer) with higher magnitude, which acts as connecting hubs with larger connectivity. The events with lower magnitude are much more but with smaller connectivity.

The analysed magnitude series is a point process, characterized by random time occurrence of the events; the point process is clusterized, because the events are neither Poissonianly nor regularly distributed on time. In order to investigate the possible dependence of the VG on the time structure of the point process, we considered the magnitude series, whose events occurring regularly (fig. 3(a)) and randomly (after random shuffling

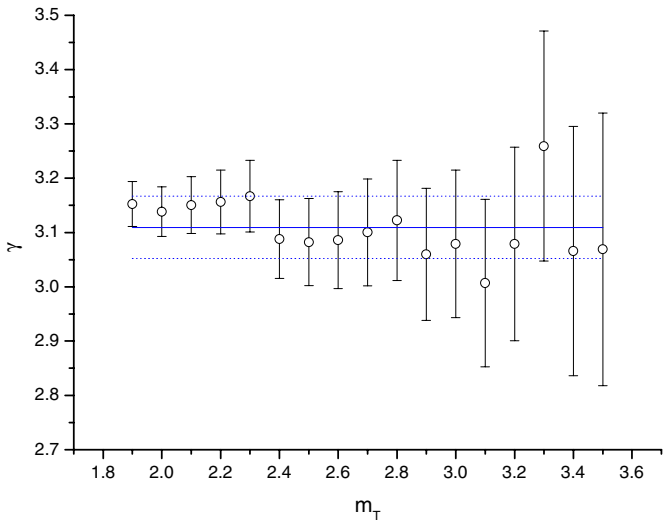

Fig. 6: (Colour on-line) Exponent of the degree distributions of the visibility graphs associated with the magnitude series for threshold magnitudes m_T ranging from 1.9 to 3.5. The exponent and the error are calculated by using the MLE method. The blue lines indicate the mean and the 1σ band.

of the interevent times) (fig. 3(b)) on time. The degree distribution for this process (red triangles in fig. 4(a)) is almost identical to that of the original sequence (black circles in fig. 4(a)), with exponent $\gamma = 3.13 \pm 0.04$ that is very close to that of the original sequence. The degree distributions for 10 randomly shuffled series are shown in fig. 4(b), along with that of the original sequence. It is clearly visible that there is no significant difference between the degree distribution of the original sequence and those of the shuffled sequences. The average exponent for the shuffled sequences is 3.18, which is almost identical to that of the original series. From this analysis, we can argue that the VG seems to depend only on the values of the magnitude.

We calculated the degree distribution increasing the threshold magnitude from 1.9 to 3.5. The results are shown in fig. 5, in which the distributions normalized to their maximum are plotted in different colours and symbols. A certain collapsing effect of all the distributions is evident. The data collapse suggests the existence of a scaling law permitting a unified description of the shape of the degree distribution, in terms of a power-law distribution.

Figure 6 shows the variation of the exponent γ (calculated by means of the MLE method) with the threshold magnitude m_T. The exponents range between 3.00 and 3.25, with mean ($\pm\sigma$) of 3.11 (± 0.06), showing that the power-law behaviour is rather stable with respect to m_T.

This study proposes a new approach of investigating earthquake magnitude series by means of the VG method. Even if more investigations should be done in the future in order to understand how the VG could be able to detect some fundamental properties of seismic processes (like the fractal distribution of the epicentres, the time clusterization, the power-law distribution of the earthquake energies, the aftershock activation, ...), such study, to our knowledge, represents the first one dealing with the application of the recent method of VG to seismicity.

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